

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

☎ (587) 879-4626 | ✉ tkuipers123@gmail.com | 🌐 tkuipers.ca | 📷 [tkuipers](#) | 🌐 [tyler-kuipers](#)

About Me

I've spent 10 years building web applications and backend systems, with the last 3 focused on architecture and technical leadership. I design APIs and data pipelines that serve multiple teams: REST APIs processing millions of events with 99.9% uptime, data flows across database models, services, and external integrations. My stack is Node.js, TypeScript, Python, and PostgreSQL on AWS. I write design documentation, lead architecture decisions, and mentor engineers. I care about building products that actually work for customers, not just shipping features. I've led teams directly and led through technical influence: RFCs, code reviews, and setting standards for a 12-person team.

Work Experience

Alberta Motor Association

SOFTWARE ARCHITECT

Edmonton, Alberta

Jan 2023 - Present

- Own 4 REST APIs serving 5 internal teams and external partners. Design API contracts, manage data flow between database models and downstream consumers, approve new endpoints, maintain OpenAPI specs. APIs process \$4.3M annually. (Python, TypeScript, REST, OpenAPI, PostgreSQL)
- Write design documentation and lead architecture decisions for 12-person team. Author RFCs, run code reviews, mentor engineers on system design and debugging. Member of Architecture Steering Committee.
- Architected event-driven system processing 2.8M events across multiple AWS accounts with 99.9% uptime. 10 Lambda functions coordinating with external partners. Won 2024 Guidewire Innovation Award. (Python, AWS Lambda, TypeScript, Terraform)
- Led modernization of document generation from legacy C# to containerized microservices. Led 2-3 engineers. Added Datadog observability for failure rates, latency, throughput. (Kubernetes, EKS, Docker, Datadog, Helm)
- Built self-service developer platform that cut environment provisioning from weeks to hours. Adopted by 15 engineers across 3 teams, created 120+ environments in 2025. (Python, Terraform, AWS, Docker)

Arterys Inc. (Acquired by Tempus Labs)

SOFTWARE DEVELOPER

Calgary, Alberta (Remote)

Aug 2021 - Aug 2022

- Built and maintained backend services for SaaS medical imaging platform. Services ran AI algorithms and returned results to the web application. Focus on low latency and reliability. (Node.js, TypeScript, AWS)
- Improved algorithm processing turnaround using Redis caching and AWS infrastructure changes.
- Monitored production via Elasticsearch and Kibana. Caught and resolved issues before they hit customers.
- Code reviews and testing for team contributions.

Alberta Motor Association

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Edmonton, Alberta

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Designed and built automated test framework from scratch that reduced defects by 30% on subsequent releases. Framework adopted org-wide, still in use 4+ years later. (Java/Gosu, JUnit)
- Led 5-6 person team using TDD. Delivered on time with no critical defects post-launch. Code reviews, mentored junior developers, collaborated with product stakeholders.
- Led Git migration for 12-15 developers. Created Docker image for local dev environments.

Pleasant Solutions

SOFTWARE DEVELOPER

Edmonton, Alberta

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD. (C#)

Lockheed Martin Canada

SOFTWARE ENGINEER ASSOCIATE

Halifax, Nova Scotia

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining real-time data streams from multiple sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system for capturing and replaying test scenarios, improving debugging and test coverage.

The King's University

SOFTWARE DEVELOPER/RESEARCH ASSISTANT

Edmonton, Alberta

May 2014 - Sep 2016

- Built web application for spatial/temporal data visualization. (JavaScript, PostgreSQL, PHP)

Education

The Open University
MSC. COMPUTING (SOFTWARE ENGINEERING)
Currently pursuing Master's in Computing at The Open University.

Milton Keynes, England (Remote)
May 2024 - Ongoing

The King's University
B.S. IN COMPUTER SCIENCE

Edmonton, Alberta
Sep 2012 - Dec 2016

Skills

**API & Data Flow
Architecture**

Design REST APIs and manage data flow across database models, API resources, and downstream consumers. Own API contracts for 5 teams, maintain OpenAPI specs, handle versioning and backward compatibility. Build event-driven systems processing millions of events.

**Technical Leadership &
Mentoring**

Lead architecture decisions for 12-person team. Write design documentation (RFCs), run code reviews, mentor engineers on system design and debugging. Member of Architecture Steering Committee. Balance technical vision with shipping products.

Scalable Systems

Build distributed systems on AWS with 99.9% uptime. Multi-account architectures, microservices on Kubernetes, event-driven pipelines. Focus on reliability, observability, and maintainability.

**Product-Minded
Engineering**

Ship end-to-end features, not just backend code. Partner with Product and Design to understand customer needs. Think about the user experience, not just the technical implementation. Troubleshoot and debug production issues.

Technologies

**Languages
Databases**

TypeScript, Python, Node.js, Java, C#, Bash
PostgreSQL, Aurora/RDS, Redis, Elasticsearch

Cloud & Infrastructure

AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM), Docker,
Kubernetes, Terraform, Helm

Architecture

REST APIs, OpenAPI, Event-Driven Systems, Microservices,
Distributed Systems

Tools & Practices

Datadog, CI/CD, Git, Agile/Scrum, TDD